FROM PASTURE TO PRODUCTION LINES

Intensification in Canadian beef vs. U.S. table-egg production

RESEARCH QUESTION Which stages of Canadian beef and U.S. egg production are intensive or extensive?

INTENSIFICATION More output • more concentrated ownership • restricted animal housing • greater automation/technology.

CANADIAN BEEF | 2 stages

MIXED SYSTEM

OUTPUT: CATTLE MARKETINGS (2011 → 2021)

52.5M → 59.9M cwt +14%

01 COW-CALF / GRAZING

EXTENSIVE

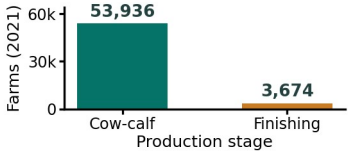
- Pasture-based, low confinement; 49.9% used rotational grazing (2021).
- Beef-cow farms: 103,673 (1996) → 61,425 (2011), −41%.
- Tech: auto-steer on 20.2% (2021); electronic cattle ID.

Calves may be sold to a different finishing owner ↓

02 FEEDLOT / FINISHING

INTENSIVE

- Fenced pens; energy-rich grain rations.
- RFID monitors individual cattle feed intake.
- Alberta ≥10k-head yards: 12 → 36 (1991–2003).



U.S. TABLE EGGS | 1 phase

INTENSIVE

OUTPUT: TABLE-EGG PRODUCTION (2012 → 2022)

80.5B → 94.0B eggs +17%

01 OWNERSHIP CONSOLIDATION

≈ 1/3 fewer large egg farms (2002–2022)

240,530 layer farms; 347 ≥100k-hen farms held 75.4% of hens (2022).

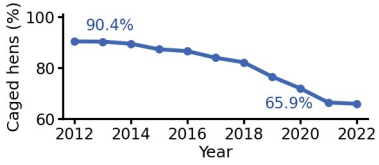
02 CONFINEMENT / HOUSING

90.4% caged (2012) → 65.9% (2022)

Cage-free hens also often remain indoors in large barns.

03 HIGH AUTOMATION

- Auto feed, water and egg conveyors.
- Computerized grading, packing and manure belts.



Farm counts track establishments, not unique owners; some beef farms combine stages.

TAKE-HOME

LIFE CYCLE: Canadian beef: extensive cow-calf grazing → intensive feedlots; U.S. commercial egg-laying: intensive.

FOUR DRIVERS: Output rose in both; ownership consolidation, confinement and automation characterize intensive phases.

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Source links remain clickable. Poster charts were drawn from the cited statistics; no additional sources were introduced.